Exploratory Data Analysis (EDA) Report – Electric Vehicle Dataset

1. Data Overview

The dataset analyzed in this study is the **Electric Vehicle dataset**, obtained from the Excel file *08_Electric Vehicle(1).xlsx*. This dataset contains information related to electric vehicle registrations and includes variables such as vehicle identification number (VIN), city, postal code, model year, manufacturer, and vehicle model. The dataset consists of **202 observations and 6 variables**, providing a structured overview of electric vehicle distribution and characteristics.

The purpose of analyzing this dataset is to better understand patterns in electric vehicle ownership, including geographic distribution, manufacturer prevalence, and model popularity. Such datasets are valuable for identifying trends in electric vehicle adoption and can support further analysis related to market growth, environmental policy, and consumer behavior. Before conducting exploratory analysis, it is essential to ensure data quality through a systematic cleaning process to improve reliability and validity of subsequent findings.

2. Data Cleaning

2.1 Missing Values

The initial assessment of the dataset identified several missing values across multiple variables. Specifically, missing entries were found in the *Vehicle Identification Number* (4 missing values), *City* (2 missing values), *Model Year* (2 missing values), *Make* (2 missing values), and *Model* (1 missing value), while the *Postal Code* variable contained no missing data.

Given that the proportion of missing values is relatively small compared to the total dataset size, listwise deletion was selected as the most appropriate data-cleaning method. This approach involves removing rows containing incomplete observations to preserve data consistency and avoid introducing bias through imputation. In particular, missing values in the VIN field are critical because VIN serves as a unique identifier for each vehicle; therefore, retaining records without this information may reduce data integrity. Similarly, incomplete entries in categorical variables such as city, manufacturer, and model could compromise subsequent descriptive and comparative analyses. As a result, rows containing missing values were removed from the dataset.

2.2 Duplicate Rows

In addition to missing values, the dataset was examined for duplicate records to prevent redundancy and overrepresentation. The analysis identified **three exact duplicate entries**, involving repeated records of electric vehicles with identical values across all variables. Examples include duplicated entries associated with Tesla Model Y, BMW i3, and Kia Niro vehicles.

To ensure analytical accuracy, these duplicate rows were removed, retaining only one instance of each repeated record. Eliminating duplicates is essential in exploratory data analysis because duplicate observations can distort summary statistics, inflate frequency counts, and bias interpretations regarding vehicle popularity or geographic concentration.

2.3 Summary of Data Cleaning

Following the data-cleaning process, the dataset was refined through the removal of incomplete and duplicated observations. These procedures enhanced the dataset's overall quality by improving consistency, accuracy, and reliability. A cleaned dataset provides a stronger foundation for subsequent exploratory data analysis, including descriptive statistics and visualization, allowing for more meaningful insights into electric vehicle trends and patterns.

3. Descriptive Statistics

Descriptive statistical analysis was conducted to summarize the primary characteristics of the cleaned electric vehicle dataset and identify meaningful patterns within the data. The analysis focused on the distribution of vehicle manufacturers (*Make*) because this variable provides insight into the relative market presence of electric vehicle brands represented in the dataset. After the data-cleaning process, the dataset contained **188 valid observations** across **23 different manufacturers**.

The descriptive statistics indicated considerable variation in the number of vehicles associated with each manufacturer. The mean number of vehicles per manufacturer was **15.67**, while the median was only **2 vehicles**, suggesting that most manufacturers contributed relatively few observations to the dataset. Furthermore, the standard deviation was **40.18**, indicating substantial variability among manufacturers. The minimum frequency recorded was **1 vehicle**, whereas the maximum reached **74 vehicles**, demonstrating a highly uneven distribution of electric vehicle brands.

Tesla was identified as the most frequently occurring manufacturer, contributing **74 vehicles**, followed by Nissan with **38 vehicles** and Kia with **19 vehicles**. Other manufacturers such as BMW and Chevrolet appeared less frequently within the dataset. These findings indicate that several dominant manufacturers account for a large proportion of the electric vehicles represented, while many other brands maintain only a minor presence.

To effectively present the descriptive statistics, a bar chart is recommended to visualize the frequency distribution of manufacturers. This form of visualization allows for clear comparison among brands and highlights the substantial dominance of specific manufacturers within the dataset.

Table 1

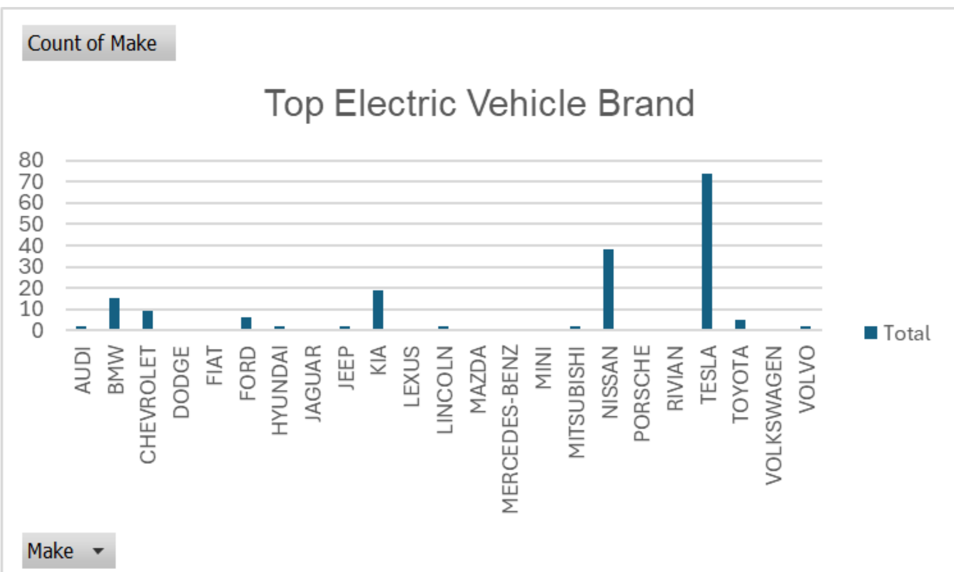
Descriptive Statistics for Vehicle Manufacturer Distribution

Statistic	Value
Number of Manufacturers	23
Total Vehicles	188
Mean	15.67
Median	2
Standard Deviation	40.18
Minimum	1
Maximum	74

Key Insights

Insight 1: Tesla Demonstrates a Dominant Market Presence

The descriptive analysis revealed that Tesla accounted for approximately **39.4%** of all vehicle records in the dataset, making it the most represented manufacturer by a substantial margin. This finding suggests that Tesla maintains a dominant position within the electric vehicle market represented in the sample. The high concentration of Tesla vehicles may reflect strong consumer preference, broader product availability, or greater market penetration compared to competing manufacturers.

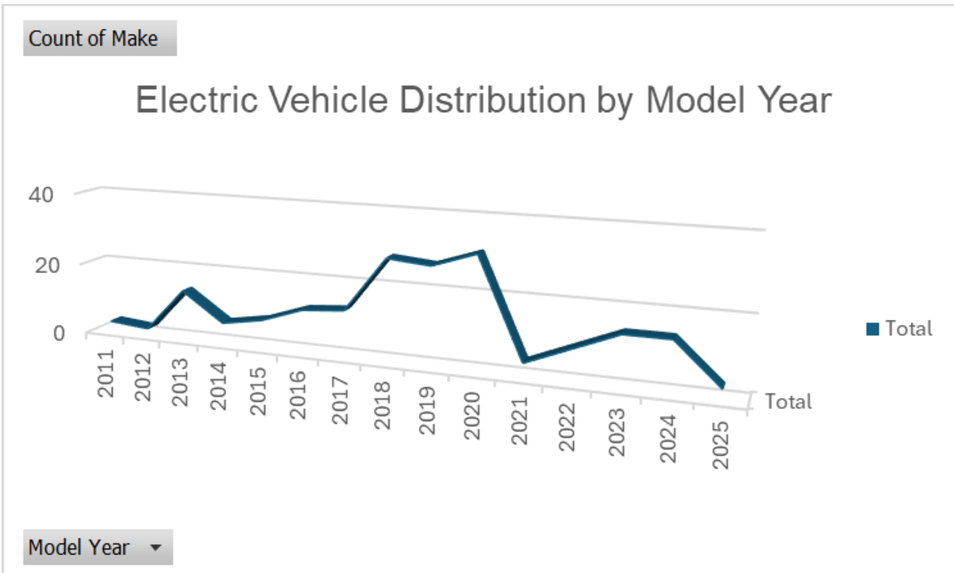


Insight 2: Electric Vehicle Adoption Increased Across Different Model Years

The descriptive analysis of the *Model Year* variable indicates that electric vehicles in the dataset are primarily concentrated in more recent years, particularly between 2018 and 2023.

This trend suggests a steady increase in electric vehicle adoption over time, reflecting growing consumer interest, technological advancements, and stronger support for sustainable transportation.

The increasing number of electric vehicles in newer model years may also indicate that manufacturers are expanding their electric vehicle offerings to meet rising market demand. This pattern highlights the rapid development of the electric vehicle industry and suggests continued future growth in electric vehicle usage and production.



Recommended Visualizations

For **Insight 1**, a **bar chart** is recommended to present the distribution of electric vehicle manufacturers. A bar chart effectively compares the number of vehicles produced by each manufacturer and clearly highlights the dominance of Tesla and other leading brands within the dataset. This visualization allows readers to quickly identify differences in manufacturer representation and better understand market concentration in the electric vehicle industry.

For **Insight 2**, a **line chart** is recommended to illustrate the distribution of electric vehicles across different model years. A line chart is appropriate because it effectively demonstrates trends and changes over time, allowing readers to observe the increasing number of electric vehicles in more recent years. This visualization helps emphasize the growth of electric vehicle adoption and the expanding development of the electric vehicle market over time.